ACCEPTED | Applicant 342011

SHAP and LIME explain the same LightGBM input and the same default-class output

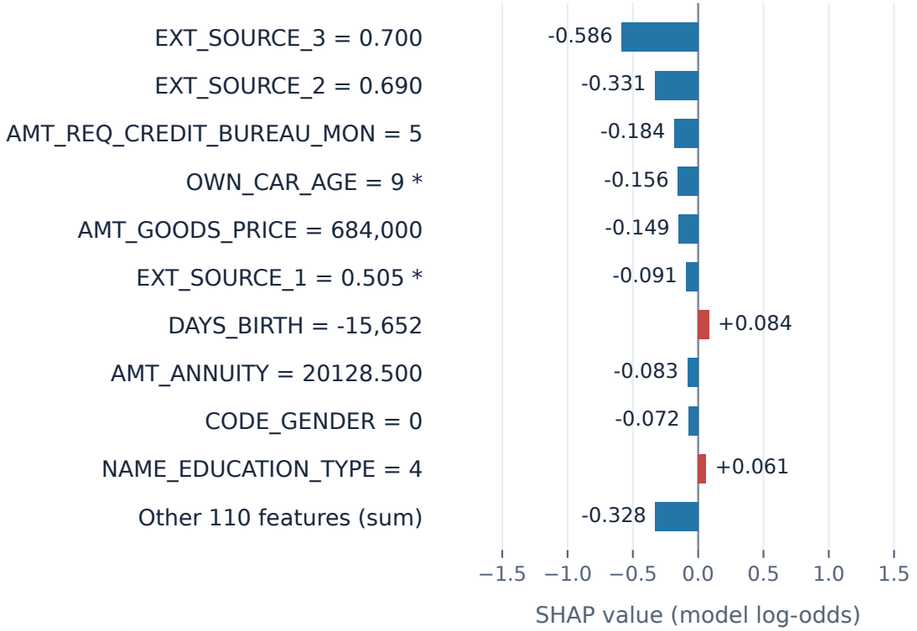
Model score 0.089

Accept < 0.500 | Reject >= 0.500

SHAP

Feature value + exact TreeSHAP attribution

Top-10 attribution coverage: 68.1% | Raw-input score: 0.141

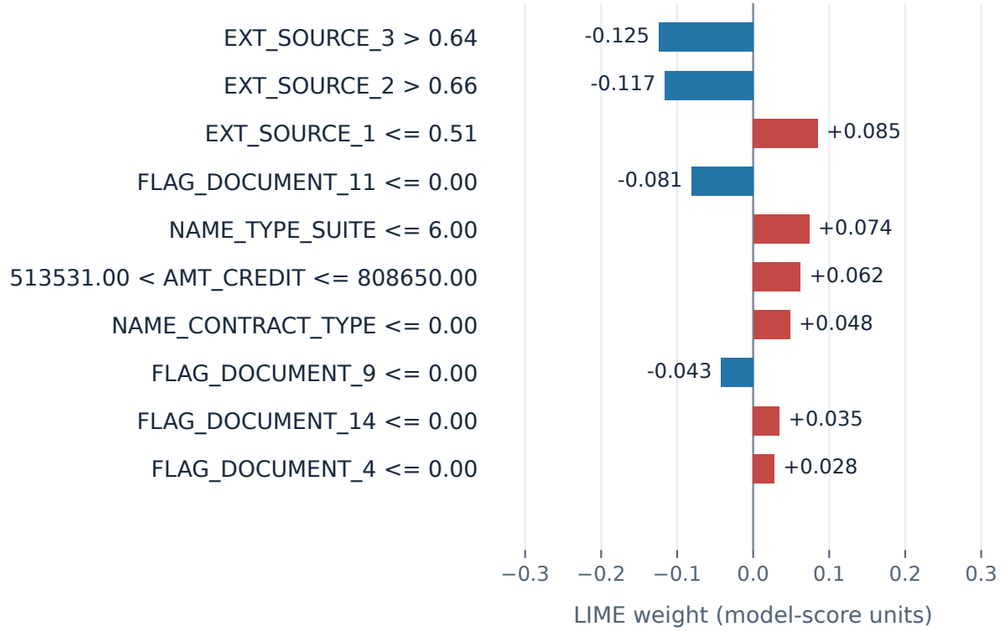


Blue / negative: lowers default-class score

LIME

Discretized condition + local surrogate weight

Local fidelity R^2: 0.293 | 1,000 perturbations | seed 42



Red / positive: raises default-class score

EXACT MODEL RECONSTRUCTION

Base -0.493 + all contributions -1.835 = -2.328

Logistic transform -> 0.089 model score

LOCAL APPROXIMATION CHECK

Surrogate 0.199 | Model 0.089 | Gap 0.110

Condition boundaries describe bins; they are not approval thresholds.

Same median-filled input for both methods; * = imputed value. Top 10 ranked within each method. Panel units differ.

Recomputed from paper code. Acceptance/rejection is a model rule, not a recorded lending decision. Score is uncalibrated.